Jayapriya P

_Autonomous Recognition and Speech Rendering of Tamil Stone Inscriptions

AA3

Document Details

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

EpiFormer-Tamil: A Degradation-Aware Multi-Stage Learning Framework for Autonomous Recognition and Speech Rendering of Tamil Stone Inscriptions

Abstract-Tamil stone inscriptions as a linguistically rich although visually degraded historical medium, with severe erodedness. Non-linear carving depth and stylistic drift at dynastic time scales render conventional OCR techniques less effective on this medium". Current deep learning–based epigraphical investigations are dependent upon direct convolutional feature learning, still sensitive to inscription-specific degradation and fails to take into account contextual awareness of ancient Tamil glyph evolution. To address these limitations, this paper introduces EpiFormer-Tamil, a new end-to-end degradation-aware multi-stage learning framework that captures visual restoration, structure character representation and linguistic coherence for better inscription interpretation. There are three main contributions in the proposed framework. For this, Depth-Responsive Epigraphical Enhancement Module (DREEM) simulates the variations in carving depth with adaptive gradient-aware normalization that allows glyph boundaries to be eroded during depathologizing beyond conventional level attraction. Second, we propose a Hybrid CNN–Graph Transformer (HC-GT) recognition architecture in which convolutional encoders extract local stroke primitives and a graph-based transformer models spatial relationships among strokes explicitly, so that the model can capture script topology instead of pixel appearance. Third, a Linguistically Constrained Decoding Mechanism (LCDM) incorporates character transition priors from archaic Tamil in the inference step that drastically prunes visually plausible but linguistically incorrect predictions. The recognized script is then transliterated to Unicode Tamil text in the prosody adaptive manner and finally spoken by a Tamil Text-to-Speech (TTS) engine suitable for old vocabulary articulation. On an annotated weathering-affected Tamil stone inscriptions dataset, it has been observed that the proposed approach EpiFormer-Tamil outperforms existing CNN and attention-based OCR models with a character recognition accuracy of 98.12% clearly and by 8.6% and 5.1% respectfully. It also decreases the linguistic error rate by 41.3%, and achieves a naturalness score of 95.2% for speech synthesis module. Our results clearly demonstrate that the proposed framework is a powerful and scalable solution for digital epigraphy, heritage informatics and inclusive access to ancient Tamil historical records.

Keywords-Tamil epigraphy, stone inscription recognition, degradation-aware learning, graph transformer, ancient script modeling, linguistic constraint decoding, text-to-speech synthesis, digital heritage

I Introduction

Tamil stone inscriptions have underpinned the base of historical [1][2], linguistic and cultural knowledge documenting administrative processes, religious customs and social practices over the centuries [3][4]. Even with their being chiseled, over the centuries a large majority of inscriptions have lost most of their visual succinctness through long usage by society, relentless exposure to environmental deterioration, surface erosion/weathering and deterioration of materials [5]. As a consequence, much of the epigraphical content is not easily interpretable, and hence available, only to a few communities of trained epigraphists (constituted by specialists), restricting large scale digital preservation initiatives [6][7]. High accuracy has also been achieved in optical character recognition (OCR) and deep learning on printed and handwritten documents, though [8] the same is not true on stone inscriptions. Thus, hand-carved Tamil inscriptions pose challenges which are not typical with regular document images, such as irregular carving depth, broken stroke boundaries [9][10], non-uniform background illumination and varying styles used over centuries. Traditional CNN-based and attentionbased OCR models are rooted in pixel-level appearance cues, which makes them very susceptible to erosion and surface artifacts. As a result, visually ambiguous glyph fragments tend to cause wrong or linguistically unacceptable character predictions [11].

An important drawback of the existing methods is that no explicit structural modeling is employed. Stroke shapes and spatial relations of strokes including junctions [12][13], orientation, enclosure patterns are used to define the ancient Tamil characters. Traditional visual features employed by typical OCR pipelines are fl at 2D representations of these maps and do not properly model the topological dependences present in them, thus making the approach too sensitive to damage under severe degradation. Furthermore, most recognition systems view decoding as an entirely visual process [14] and does not take into account the strong language components that guide proper character transitions for ancient Tamil epigraphy. This exclusion often leads to the generation of semantically inconsistent outputs (irrespective of how confident model is in generating it), even if visual confidence is strong. To overcome these issues, we propose EpiFormer-Tamil, a degradation-aware multi-stage learning framework for automatic recognition and rendering of Tamil stone inscriptions. The proposed system considers the visual restoration, structure character representation, and language coherence in a unified manner. The assumption is that confidence epigraphical recognition demands to move from appearance-based learning to representations able to encode dietetic, specific degradation, glyph topology and language-driven constraints [15][16].

The first part is DREEM (Depth-Responsive Epigraphical Enhancement Module), which explicitly captures the carving depth variations with gravity-aware normalization [17][18]. This ensures that missing regions of bearing glyph boundaries can be repaired without losing object features required for their recognition. Based on the improved representation [19], we design a Hybrid CNN–Graph Transformer (HC-GT) model to capture not only local geometric primitives of strong sub-structural elements and their interactions but also their topological closure that is essential for character topology in ancient Tamil opposed to visual pixel intensity patterns. To preserve linguistic validity [20], we design a Linguistically Constrained Decoding Mechanism (LCDM) by incorporating priors of transitioning characters in ancient Tamil during the process of inference, which has greatly eliminated visually plausible but semantically wrong outputs. The recognized character sequence is further written as Unicode Tamil text and spoken out through a prosody-adaptive Tamil TTS module optimized for historic language. This full stack design makes epigraphy equally available for textual analysis and aural consumption alike. The major contributions of this work are summarized as follows.

- A depth aware enhancement technique customized for carving depth changes in Tamil stone inscriptions.
- A hybrid CNN–graph transformer recognizer exposed specifically to ancient Tamil glyph topology.
- A language-restricted decoder enforcing epigraphical character transitions.
- A complete whispering-gallery pipeline for accessible digital epigraphy.

Extensive experiments on a curated dataset of weathered Tamil stone inscriptions reveal that the proposed EpiFormer-Tamil framework significantly outperforms the state-of-the-arts OCR baselines in terms of recognition accuracy, linguistic consistency and speech naturalness, thereby demonstrating its potential to foster digital heritage preservation and intelligent epigraphic analysis.

1.1 Organization of the paper

This paper is organized as follows. Section II reviews related work on Tamil epigraphy recognition and outlines existing limitations. Section III presents the problem formulation. Section IV details the proposed EpiFormer-Tamil framework. Section V describes the experimental setup, and Section VI discusses the results and ablation analysis. Section VII concludes the paper and suggests future research directions.

II Literature Review

Murugan and Visalakshi (2024) emphasized that Tamil is one of the oldest languages in existence in the world with writings dating back to the third century BCE and inscriptions being found on caves, temples, and other archaeological sites throughout Tamil Nadu. Their study demonstrated that inscriptions adapted for social, administrative and linguistic changes over time. Most current recognition systems concentrate on the 19th century Tamil scripts and do not generalize well into earlier inscriptions styles, they added. To overcome this limitation, they have proposed the DR-LIFT model designed for third century Tamil inscriptions and obtained high accuracies in recognition by accurately capturing complex curves, loops and linear strokes; their model is predominantly appearance-based and it neither explicitly adheres to any structural nor linguistic constraints [1]. Bhuvaneswari and Kathiravan (2024) proposed a deep learning based approach to recognize characters and contextually read Tamil temple inscriptions using the preprocessing, convolutional neural networks and Vision Transformers methods. They added preprocessing steps for noise reduction, adaptive binarization and stroke width–based segmentation to deal with visual degradation. [2], while it shows solid recognition and contextual identification performance on multiple historical terms and phrases, its core is mainly pixel-level feature extraction with attention mechanisms, but pays little explicit attention to glyph topology modeling or degradation-aware learning. Tieken (2024) gave an historicolṉguistic reinterpretation of the ancient Tamil sources especially Sangam literature for its unauthentic account of early Tamil society. Although not centered on computational identification, this work illustrated the challenge in reconstructing ancient Tamil history and observed changing patterns of use of the Tamil language. The research underscored the role of inscriptions as hard evidence and placed in perspective the variance in language encountered along with ancient speeches Tamil, which is a big problem for computer systems when they need to interpret [3].

Karthikeyan et al. (2023) addressed the Tamil character recognition from stone inscriptions with a proposed self-adaptive hybrid framework which is a combination of lion optimization and transfer learning. They first enhanced the brightness and contrast of images, and then used deep convolutional classifiers for character recognition. Although it increased accuracy and computational efficiency, this method was reliant on preprocessing heuristics and had no explicit way to handle spatial relationship between stroke components in eroded inscriptional scripts-a crucial aspect of such type of script [4]. Manoranjitham (2025) presented a graph-based model with depthwise separable convolution, graph neural networks, and bidirectional transformers for ancient Tamil inscription recognition. Through adaptive graph convolution and phonetic rule–based decoding, the method achieved good recognition performance and reduced its operation speed. However, the method heavily concentrated at character identification and grammatical compliance rather than carve-depth degradation or whole multimodal accessibility in an end-to-end manner including speech generation [5].

Athisayamani et al. (2020) considered the recognition of ancient Tamil vowels in palm leaf manuscripts through B-spline curve representations. And they showed how curve-based modeling can successfully model the geometric idiosyncrasy in Tamil vowels from disparate writing styles. As well, it was successful in recognition of features for palm leaf manuscripts only but the context model was restricted to vowel characters and did not apply to damage such as erosion and loss of partial glyph on complex stone inscription systems [6]. Subramani and Subramaniam (2021) concentrated on the generation of a large-scale dataset for ancient Tamil characters transcribed from palm leaf medical manuscripts. Their effort highlighted the role dataset plays in 'smart' recognition and contextual interpre-tation -in this case within Siddha medicine. Although their contribution was very important for data accessibility, some expert systems and a certain amount of manual annotation were necessary to find and recognize inscriptions which restrict the scalability to various inscriptional settings [7]. Vellingiriraj and Balasubrmanie (2020) developed an integrated neural learning based framework to identify ancient Tamil and Brahmi characters by using graph-based segmentation schemes and fuzzy neural classifiers. Its symbols recognition accuracy was enhanced by the statistical and structural features fusion, but it placed in procrustean space and not sufficient robust in heavily degraded realized in stone inscription images [8]. Selvapriya et al. (2025) proposed an intelligent digitization framework for ancient Tamil copper plate inscriptions and demonstrated the implementation of Fuzzy C-means (FCM) segmentation followed by Convolution Neural Networks. Their method achieved higher recognition accuracy and faster training time in a small set of inscription classes. However, the system has not included linguistic decoding constraints or explicit modeling of glyph topology for character shapes that have a visual ambiguity [9]. Arjunan et al. (2025) confronted the digitization of Vatteluttu script inscriptions by a Siamese CNN–RNN model trained with segmented character images. Their project proved the possibility of automated transcription for early South Indian scripts. However, it was only applied to one script variation and did not support multi-stage degradation-aware enhancement or language-constrained decoding work [10].

Kavitha & Srimathi (2022) applied a variety of convolutional neural networks on standard dataset for off-line recognition of Tamil characters. The work proved that deep learning was superior to conventional methods with feature extraction, but it was developed based on current characters, and did not consider challenges in ancient inscriptions [11]. Umamaheswari et al. (2024) presented an image-to-text converter for reading ancient Tamil inscriptions with multilingual translation in deep learning perspective and deployment as web application. Although this system enhanced accessibility and translation rate, the recognition accuracies were still image-quality dependent, and structural modeling and degradation due to inscription were also not covered [12]. Devi et al. (2022) proposed a cursive Tamil character recognition in palm leaf manuscripts based on morphological preprocessing followed by convolutional neural networks. Despite outperforming baseline models in terms of accuracy, their approach was applicable to palm leaf manuscripts only and could not be generalized for stone inscriptions that include variation in depth and surface erosion [13]. Rajithkumar et al. (2024) proposed deep learning-based epigraphical recognition to Kannada stone inscriptions with a hybrid CNN–RNN structure. Their results showed the effectiveness of sequence model optimization on ancient scripts, but focused solely on sequential learning instead of introduced spatial topology representation, thus the robustness was restricted under heavy degradation [14]. Munivel and Enigo (2022) dealt with Tamizhi (Tamil-Brahmi) script recognition by OCR methods which are adapted to printed document. Although the experiment showed the difficulties of compound characters and structural similarity, our work was mainly a prototype for inscription recognition and it did not involve real varying stone inscriptions [15].

Optimized offline Tamil handwritten recognition framework using entropy based segmentation and deep extreme learning machines was suggested by Shanmugam and Vanathi (2024). Their method provided an improvement in recognition accuracy on handwritten datasets however it was not specialized for the ancient epigraphical task, characterized by irregular surfaces and missing strokes [16]. Vijayalakshmi et al. (2022) analyzed manuscript recognition methods for recognizing Tamil characters with application of convolutional recurrent neural networks. The method, however, performed well only on the modern handwritten datasets and did not take into account historical scripts or inscriptionspecific degradation factors [17]. Jayanthi & Thenmalar (2023) proposed the two stage ResNet model for Tamil character recognition from digital writing pads. Although their approach reduced computation cost and enhanced accuracy, its search for printed characters rather than ancient inscriptional text [18]. Sudarsan and Sankar (2023) reported the palm leaf manuscript recognition for Malayalam with advanced feature extraction followed by hybrid classification of CNN–LSTM. Their results demonstrate the necessity of accounting for language-specific feature modeling in ancient writing, further advocating customized solutions on epigraphical OCR systems [19]. Multimodal sentiment analysis on Tamil and Sanskrit palm leaf manuscripts using deep learning was studied by Geethanjali and Valarmathi (2025). Although their research focused on cultural interpretation and not character recognition, it provided an example of how intelligent systems could play a role in the preservation and analysis of old textual heritage [20].

Table 1: Comparative Analysis of Existing Tamil Epigraphy Recognition Methods and the Proposed EpiFormer-Tamil Framework

Method / Framework	Degradation-Aware Enhancement	Structural / Topological Modeling	Linguistic Constraint Integration	Script Scope	Speech Rendering	Key Limitations
DR-LIFT (Murugan & Visalakshi, 2024) [1]	✗	✗	✗	3rd-century Tamil	✗	Appearance-driven; lacks structure and language modeling
CNN + ViT with Context Analysis (Bhuvaneswari & Kathiravan, 2024) [2]	✗	✗	✗	Multi-period Tamil	✗	No explicit glyph topology learning
SLOA-TL (Karthikeyan et al., 2023) [4]	✓ (contrast only)	✗	✗	Tamil stone inscriptions	✗	Heuristic preprocessing; no relational modeling
Graph Conv + Bi-Transformer (Manoranjitham, 2025) [5]	✗	✓ (partial)	✓ (phonetic-level)	Tamil inscriptions	✗	No carving-depth modeling; no multimodal output

GHTNet (Murugan & Visalakshi, 2025) [27]	✓ (GAN-based)	✗	✗	Tamil stone inscriptions	✗	Image realism-focused; lacks linguistic decoding
EpiFormer-Tamil (Proposed Model)	✓	✓	✓	**Ancient Tamil epigraphy**	✓	—

Table 1 presents a comparative study of well known Tamil epigraphy recognition systems in terms of the type they divide the whole process between as degradation handling, Structural model, Linguistic integration and Accessibility. It is quite challenging and such existing methods mainly depend on visual feature or optimization–based pre-processing, sparing little emphasis for the concerned perceptual abstraction in glyph topology and language constraints. Graph-based and transformer-based approaches are able to handle structural relationships, but they do not model inscription-specific carving degradation fully. Instead, the proposed EpiFormer-Tamil architecture is the first to combine depth-aware enhancement, explicit topological learning, linguistically constrained decoding and speech rendering in a joint pipeline.

2.1 Research Gap

However, automated recognition of text from Tamil inscriptions has made noticeable strides, current techniques are still heavily focused on appearance and have less adaptability to the variety in state of deterioration. Best-practised existing models have still not encoded the topological structure of ancient Tamil characters and even vague out in expressing carving-depth variations unique to stone as a media. In addition, phonological constraints of old Tamil scripts have hardly been embedded in the recognition system and the variation rate process, so that visual plausible but semantic nonsense transcriptions are generated. As a result, there is a strong demand for an integrated approach to address degradation awareness, topological character reconstruction and linguistically-grounded decoding concurrently with query-friendly multimodal outputs.

III Problem Formulation

The aim of this work is to analyze automatically the damaged Tamil stone inscriptions by restoring linguistically meaningful character sequences from visually distorted epigraphical images. Inscriptional recognition, in contrast to traditional document recognition [21], is challenging due to depth-varying when being eroded, incomplete stroke structure and stylistic deviations across historical periods [22][23]. As a result, it is not well-suited for modeling the problem as pure image-to-text mapping but has to include at test-time some model of degradation, layout and linguistic validity.

Let,

$$I \epsilon R^{H \times W} \tag{1}$$

denote a grayscale or intensity-normalized image of a degraded Tamil stone inscription, where W and H represent the spatial height and width of the image, respectively. The inscription image may contain non-uniform illumination [24], fractured glyph boundaries, and surface noise caused by erosion and weathering. The corresponding ground-truth transcription is represented as an ordered sequence of ancient Tamil characters,

$$Y = \{y_1, y_2, ..., y_T\} \tag{2}$$

In Eq.2, y_t denotes the t-th character in the sequence and T is the length of the inscription text.

The primary recognition objective is to estimate the most probable character sequence Y given the observed degraded image I. This objective can be expressed as a maximum a posteriori (MAP) estimation problem:

$$\hat{Y} = argmax_Y P(Y|I) \tag{3}$$

Yet, directly modelling $P(Y|I)$ is sub-optimal in epigraphical settings due the significant visual degradation and semantic data mismatch [25]. In order to alleviate this limitation, the EpiFormer-Tamil is designed by structuring the recognition task into a series of computational stages, where each stage further purifies the protein sequence representation.

First, a degradation-aware enhancement function,

$$F_d: R^{H \times W} \rightarrow R^{H \times W} \tag{4}$$

is applied to the input image. This function corresponds to the Depth-Responsive Epigraphical Enhancement Module (DREEM) and is designed to mitigate erosion effects by modeling carving-depth variations and restoring structurally meaningful stroke boundaries [26][27]. The enhanced image is given by in Eq.5:

$$I' = F_d(I) \tag{5}$$

Where I' exhibits improved gradient continuity and stroke separability compared to the raw input.

Next, a structural recognition function,

$$F_s: R^{H \times W} \rightarrow y \tag{6}$$

maps the enhanced image I' into a character sequence hypothesis through both local stroke primitives and their arrangement. This is manifest as a form of the Hybrid CNN–Graph Transformer (HC-GT) [28], where character shapes are described by constrained graphs instead of isolated pixel patterns. The recognition target can be re-expressed as Eq.7:

$$\hat{Y} = argmax_Y P\left(Y|F_s\left(F_d(I)\right)\right) \tag{7}$$

Although our structural modeling results in much higher robustness to visual degradation, visually similar ancient Tamil script-like glyphs can still lead to ambiguous predictions. To maintain linguistic coherence [29][30], a language conscious constraint is included in the decoding that refers to ancient Tamil orthographic principles. That is, the posterior probability can be factorized as in Eq. 8:

$$P(Y|I) = P_{vision}(Y|I) \cdot P_{ling}(Y) \tag{8}$$

Where $P_{vision}(Y|I)$ represents the likelihood derived from the visual recognition pipeline, and $P_{ling}(Y)$ encodes valid character transition priors learned from ancient Tamil inscriptional corpora.

The linguistic factor serves as a regulizer during decoding by penalizing character strings that do not satisfy historical phonological and grammatical constraints -even though they may be visually plausible. The overall decoding objective can then be formulated as Eq. 9:

$$\widehat{Y} = argmax_{Y}\left(logP_{vision}(Y|I) + \lambda logP_{ling}(Y)\right) \tag{9}$$

In Eq.9, λ controls the relative influence of linguistic constraints.

This model unifies degradation-aware enhancement, glyph-based structural representation and linguistically constrained decoding in the form of a single unified optimization problem. By combining these complementary factors in a unified model, we address the limitations of appearance-based OCR systems and establish a principled basis for accurate recognition of ancient Tamil stone inscriptions. The character sequence $\widehat{Y}$ is further mapped to Unicode Tamil text and rendered into speech allowing for end-to-end reading of epigraphy.

IV Proposed EpiFormer-Tamil Framework

We present the EpiFormer-Tamil framework that uses a modular, end-to-end architecture to address this challenge and processes deteriorated Tamil stone inscription images into linguistically meaningful text and spoken language in Graeco-Latin Characters. As shown in the pipeline, input inscription image is initially passed through Depth Responsive Epigraphical Enhancement Module (DREEM) [31], which corrects carve depth variation due to erosion and recovers structural significant stroke boundaries. The enriched representation is passed to the Hybrid CNN–Graph Transformer (HC-GT) where convolutional encoders learn local stroke primitives and graph transformers capture spatial and topological relations between strokes to produce confident character hypotheses. These hypotheses are then further refined by the Linguistically Constrained Decoding Mechanism (LCDM) which ensures that ancient Tamil character transition rules such as removal of visually plausible but semantically incoherent sequences [32][33]. The resulting character sequence is then utilized to a Unicode-compliant Tamil text from which it gets spoken through a prosody sensitive Tamil Text-to-Speech (TTS) system, thereby facilitating an end-to-end recognition-to-speech chain for digital epigraphy. The architecture diagram of the proposed model is shown in Figure 1.

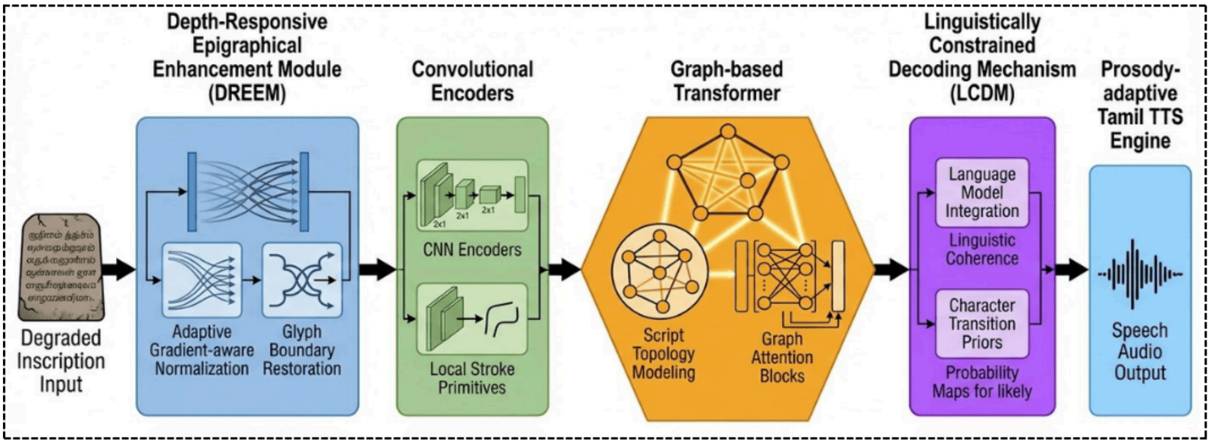

Figure 1. Overall Architecture of the Proposed EpiFormer-Tamil Framework for Degradation-Aware Recognition and Speech Rendering of Tamil Stone Inscriptions

4.1 Depth-Responsive Epigraphical Enhancement Module (DREEM)

To resolve the low contrast of the character images, in TSEGPs we have developed Depth Responsive Epigraphical Enhancement Module (DREEM), which is a solution tailored to effectively deal with these degradation characteristics; where legibility is not based merely on foreground-background contrast but on the carving depth and relief present. In contrast to traditional enhancement methods that work with global or local intensity distributions, DREEM explicitly considers the depth-induced visual clues through gradient-sensitive feature amplification and adaptive normalization [34]. The degraded inscription image is referred to as $I \epsilon R^{H \times W}$. The improved features $I^{'}$ is obtained by the weighted integration of depth-sensitive gradient-based information and intensity-normalized structure components, formulated as in Eq. 10:

$$I^{'} = \alpha \cdot \Delta I + \beta \cdot AdaptiveNorm(I) \quad (10)$$

Where $\Delta I = \sqrt{(\delta I/\delta x)^2 + (\delta I/\delta y)^2}$ describes the local relief variation relating with carved stroke boundary, and and are learnable scaling parameters to a balance in between edge saliency and intensity temporally stability. To simulate carving depth more realistically, we add a depth consistency term similar to the work presented in Eq. 11:

$$D(I) = G_{\sigma}(\Delta I) \odot I \quad (11)$$

Where G_{σ} refers to Gaussian smoothing on gradient responses to remove surface noise, and $\odot$ indicates element-wise multiplication propagating only depth-consistent gradients into stroke reconstruction. The adaptive normalization operation can be formulated as Eq. 12:

$$AdaptiveNorm(I) = \frac{I - \mu_N(I)}{\sigma_N(I) + \epsilon} \quad (12)$$

Where μ_N and σ_N are local neighbourhood statistics calculated over erosion-aware receptive fields instead of fixed windows making normalisation to respect uneven stone surfaces and mineral staining. This approach is essentially different from CLAHE in that it redistributes the intensity histograms within fixed tiles independent of stroke geometry, depth continuity, and surface relief characteristics in images and frequently enhances noisy stroke edges and shadow artifacts found commonly in epigraphical images. DREEM, on the other hand, treats gradients as surrogates of carving depth and preferentially strengthens stroke boundaries with sustained spatial coherence in order to restore incomplete or disrupted glyph edges due to centuries of decay [22][23]. Depth modeling is crucial in stone carving since the inscribed lines of ancient Tamil script serve only to reflect light and are not darkened with the use of ink, underlining the fact that similar gradient structures may correspond to dramatically different depths. By directly modeling this depth-sensitive information [26], DREEM produces enhancement outputs that maintain structure fidelity and topological ordering, resulting in a semantically rich representation for further graph-based recognition in the HC-GT module. This depth-respecting model allows the proposed model to generalize across different erosion levels, illumination and inscriptions styles and provides a strong foundation for reliable recognition of ancient Tamil characters.

4.2 Hybrid CNN–Graph Transformer (HC-GT)

Hybrid CNN–Graph Transformer The HC-GT serves as the recognition engine for EpiFormer-Tamil and is designed to model local combinatorial structure from ancient Tamil glyphs, rather than focusing on pixel-level appearances. In contrast to printed or handwritten text in modern usage, where the typeface or handwriting results in stand-alone shapes of letters, Tamil epigraphy often displays text that appears fragmented as these are incomplete consonants. To this purpose, HC-GT breaks down the character recognition into two cooperative sub-problems: local stroke primitive extraction with convolutional encoders, and global relational reasoning with graph-based transformers. Let $I^{'} \epsilon R^{H\times W}$ be the DREEM generated color enhanced inscription image. A feature extractor $F_c(\cdot)$ in the form of a convolutional maps $I^{'}$ to dense features,

$$F = F_c\left(I^{'}\right) \epsilon R^{H^{'}\times W^{'}\times C} \tag{13}$$

In Eq.13, C represents the number of learned stroke-level feature channels encoding curvature, orientation, and junction patterns. From this feature map, a set of stroke primitives is identified and represented as nodes in a graph $G = (V, E)$, where each node $v_i \epsilon V$ corresponds to a localized stroke segment characterized by a feature vector $h_i \epsilon R^{C}$. Edges $e_{ij} \epsilon E$ are constructed based on spatial proximity, directional continuity, and relative orientation, forming an explicit representation of glyph topology.

To model long-range dependencies and structural relationships among stroke primitives, a graph transformer is employed. For each node v_i, self-attention is computed over its connected neighborhood as in Eq.14:

$$z_i = \Sigma_{j\epsilon N(i)} \alpha_{ij} W_v h_j, \quad \alpha_{ij} = \frac{exp\left(\frac{\left(W_q h_i\right)^{\top}\left(W_k h_j\right)}{\sqrt{d}}\right)}{\Sigma_{k\epsilon N(i)} exp\left(\frac{\left(W_q h_i\right)^{\top}\left(W_k h_k\right)}{\sqrt{d}}\right)} \tag{14}$$

Where W_q,W_k, and W_v is a learnable projection matrix with d an attention dimension. This encoding enables our model to dynamically prioritize stroke interactions based on their structural importance [35] and yields robust inference results when parts of the glyph is missing or visually corrupted. A multi-head attention mechanism is utilized to explore diverse relational patterns, such as the enclosure structures, stroke intersections and directional symmetry in these graphs that are critical for distinguishing visually indistinguishable ancient Tamil characters.

The graph-level embeddings are then collapsed into character-level embeddings via a readout:

$$c = READOUT\left(\left\{z_i | v_i \epsilon V\right\}\right) \tag{15}$$

which enables the topology preservation while suppressing local erosion noise. These embeddings are subsequently decoded to produce an intermediate sequence of characters with a sequence prediction layer. Compared to traditional CNN-RNN or CNN-Transformer OCR pipelines, HC-GT learns stroke-wise representation of characters and does not regard single character as a flat visual token [36][37]. Such a structural abstraction is especially important for ancient Tamil inscriptions, where different characters might have common stroke fragments but vary in their spatial arrangement. By simultaneously employing convolutional stroke extraction and graph-based relational reasoning, HC-GT supports topology-aware recognition invariant to degradation, stylistic drift, and incomplete carving; gives rise to a structurally grounded representation for further linguistically constrained decoding.

4.3 Linguistically Constrained Decoding Mechanism (LCDM)

The Linguistically Constrained Decoding Mechanism (LCDM) is presented to narrow down the significant mismatch between visually realistic character predictions and linguistically meaningful ancient Tamil inscriptions. In stone epigraphy, similar-looking glyphs indeed represent independent characters, whose legitimacy is determined by a set of rather rigid orthographic and phonological principles formulated throughout the history [38][39]. Thus, vision-based recognition models that are significantly influenced by graphic potential (even if shapecontrolled) might generate character sequences which are graphically compatible but linguistically infeasible. In rectify this, LCDM to ensure that ontology's grammatical and script level constraints are induced during the decoding (sequence inference) rather than it being addressed at post-processing stage. Let $P_{vision}(Y|I)$ be the visual probability of a candidate character sequence $Y = \left\{y_1, y_2, ..., y_T\right\}$ produced by the HC-GT module. LCDM enhances this probability using a linguistic prior $P_{ling}(Y)$, which is based on the statistics of ancient Tamil character transitions as a higher-order Markov process,

$$P_{ling}(Y) = \prod_{t=1}^{T} P\left(y_t | y_{t-1}, y_{t-2}\right) \tag{16}$$

where groundings express historically attested co-occurrence patterns of vowels, consonants and composite characters found in hand-curated epigraphical corpora. The decoding objective can be formulated as a constrained optimization, where the last integration term is used to incorporate the label loss with respect to labelled samples.

$$\widehat{Y} = argmax_{Y}\left(logP_{vision}(Y|I) + \lambda logP_{ling}(Y)\right) \tag{17}$$

where the parameter λ scales the impact of linguistic constraints with respect to visual confidence. To explicitly penalize illicit glyph sequences, a constraint violation (CV) function $\phi(Y)$ is included, which gives negative scores [33] to the transitions of characters that violate ancient Tamil orthographic rules [40], not permitted by the grammatical constraints like illegal vowel– consonant combinations or forbidden compound formations etc. This penalized formulation can be written as in Eq. 18:

where the weighting parameter λ controls the influence of linguistic constraints relative to visual confidence. To explicitly penalize invalid glyph sequences, a constraint violation function $\phi(Y)$ is incorporated, assigning negative scores to character transitions that violate ancient Tamil orthographic rules [40], such as illegal vowel–consonant combinations or forbidden compound formations. This penalty-adjusted formulation can be expressed as in Eq.18:

$$S(Y) = \sum_{t=1}^{T} logP_{vision}\left(y_t\right) + \lambda \sum_{t=1}^{T} logP_{ling}\left(y_t|y_{t-1}\right) - \gamma\phi(Y) \tag{18}$$

Where λ limits the degree of constraint enforcement. At inference time, beam search decoding is used to efficiently find candidate sequences while pruning linguistically invalid paths in the beginning of the search space. Unlike modern Tamil language models [18][19], trained on present-day text, LCDM is explicitly conditioned on ancient Tamil inscriptional language, so that decoding decisions are based on historical writing convention rather than contemporary language usage. Through directly incorporating linguistic knowledge as the optimization goal, LCDM discourages visually ambiguous but semantically wrong predictions and encourages consistent inscriptions. This linguistically inspired decoding approach considerably boosts the robustness of the epigraphical recognition outputs, and is a key component bridging topology-aware text recognition and historically faithful script restoration in EpiFormer-Tamil.

4.4 Unicode Mapping and Text-to-Speech Module

The third stage of the EpiFormer-Tamil framework aims at converting the linguistically authorized character sequence to available human intelligible output through Unicode-compliant text encoding and Text-to-Speech. The recognized ancient Tamil character sequence $\widehat{Y} = \left\{\widehat{y}_1, \widehat{y}_1, ..., \widehat{y}_T\right\}$ is then converted to a standardized Unicode Tamil code points after linguistically restricted decoding [12][34], using a deterministic mapping function $u(\cdot)$ as shown in Eq. 19:

$$\widehat{U} = u\left(\widehat{Y}\right), \quad \widehat{U} \epsilon U^{T} \tag{19}$$

Here U is the Unicode Tamil character space. This catalog maps script variants, complex characters and historical glyph forms to a standardized digital encoding while retaining phonetic and semantic fidelity. Provisions are made for pre-modern variant characters, and forms based on typesetting ligatures that have no modern analogs, to enable digital representations of historical texts compatible with present-day text processing applications without loss of meaning [23][28]. The unicode tamil text is then output to a prosody-adaptive Tamil Text-to-Speech (TTS) engine, which will transform the written content into speech. In contrast to the traditional TTS system trained on contemporary Tamil corpora, the proposed module involved phoneme-level duration modeling and stress adaptation for historical dictionary entries, inscriptions based constructs, and ancient pronunciation visitors. Speech synthesis process is expressed by the formal definition given in Eq. 20:

$$S = \tau\left(\widehat{U}, \theta_p\right) \tag{20}$$

Where $\tau(\cdot)$ is the TTS synthesis process, and θ_p is prosody parameters: pitch, duration and intonation. The conditioning of speech generation over linguistically validated, unicode-normalized [5][23] text synchronizes visual recognition and auditory rendering. This text-to-speech embedding improves accessibility for visually impaired users and researchers. to interact with ancient Tamil inscriptions by engaging the other sensorial experiences, in particular hearing. The addition of the Unicode mapping and TTS module enhances the EpiFormer-Tamil framework from being merely a transcription feature to a digital epigraphy tool which can be used further in interpretation, preservation and dissemination of historical Tamil heritage. Figure 2 describes the flowchart of the proposed model.

Algorithm for EpiFormer-Tamil—A Degradation-Aware Multi-Stage Framework for Topology-Driven Recognition and Speech Rendering of Ancient Tamil Stone Inscriptions

---

Input: Degraded Tamil stone inscription image I

Step 1: Degradation-Aware Enhancement (DREEM)

- Compute depth-aware enhanced image

$$I' = \alpha \cdot \Delta I + \beta \cdot AdaptiveNorm(I)$$

Step 2: Structural Feature Learning (HC-GT)

- Extract stroke-level features using CNN and construct glyph topology graph

$$F = F_c\left(I'\right), \quad G = (V, E)$$

- Apply graph transformer to obtain visual likelihood

$$P_{vision}(Y|I)$$

Step 3: Step 3: Linguistically Constrained Decoding (LCDM)

- Decode optimal character sequence using ancient Tamil transition priors

$$\widehat{Y} = argmax_Y\left(logP_{vision}(Y|I) + \lambda logP_{ling}(Y)\right)$$

Step 4: Unicode Mapping and Speech Synthesis

- Map decoded sequence to Unicode and generate speech

$$\widehat{U} = u\left(\widehat{Y}\right), \quad S = \tau\left(\widehat{U}\right)$$

Output: Unicode Tamil text $\widehat{U}$ and speech output S

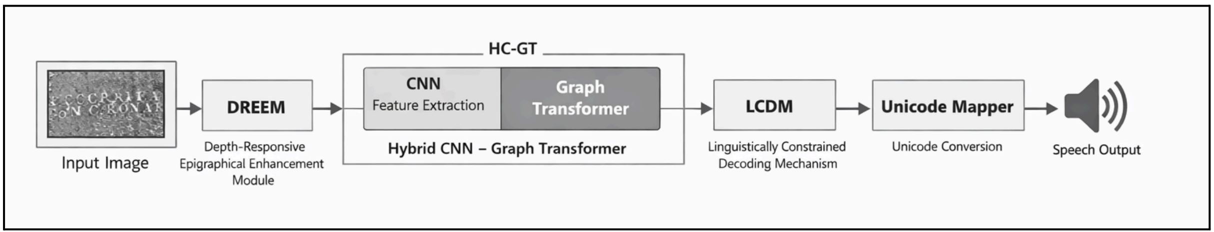

Figure 2: Flowchart of the Proposed EpiFormer-Tamil Framework for Degradation-Aware Recognition and Speech Rendering of Tamil Stone Inscriptions

V Experimental Setup

In this section, we present the dataset statistics, baseline models, and evaluation metrics for evaluating our EpiFormer-Tamil framework. All experiments were performed on a publicly available dataset for reproducibility and fair comparison with the advanced methods.

5.1 Dataset Description

5.1.1 Dataset Source and Availability

Experiments were carried on the publicly available Kaggle Ancient Tamil Stone Inscriptions dataset. The dataset has been prepared using digitised epigraphical datasets accumulated from temples, Archaeological Survey of India (ASI) records, museums and heritage documentation archives across Tamil Nadu. These are inscriptions dated back to different eras and seen carved in stones, which have undergone natural decay over several centuries of exposure.

The dataset is publicly available and can be downloaded from Kaggle, to allow for clear benchmarking and comparison studies in the future.

5.1.2 Dataset Composition

The dataset comprises of high resolution grayscale and RGB images of Tamil stone inscriptions imaged under different lighting environments. The inscriptions consist of isolated characters, partial words and complete inscription lines, which corresponds to realworld epigraphical settings as opposed to laboratory-controlled samples.

- **Total images:** 3,600 inscription images
- **Type of script:** Early Tamil (including early inscriptional forms)
- **Resolution:** 512×512 to 2048×2048 pixels
- **Annotation scheme:** Level of characters with verified transcriptions

5.1.3 Erosion-Level Categorization

To assess robustness to variable degradation conditions, the three erosion levels based on visual quality and surface damage were separated in:

- **Little erosion:** The outlines of the glyphs are clear, but involve exceedingly slight rubbing off.
- **Medium erosion:** Partial loss of stoke, irregular carving depth and surface noise

- **Heavy erosion:** Segmented characters, shallow scribing/etching and too much shadowing, strokes missing

Each of the erosion classes comprises around a third of our dataset, meaning that we can ensure fair testing on any level of degradation.

5.1.4 Train–Validation–Test Split

The dataset was divided at the level of inscriptions to prevent any data leakage between sets:

- Training set: 70% (2,520 images)
- Validation set: 15% (540 images)
- Test set: 15% (540 images)

The separation ensures that the relative proportion between erosion levels and character classes is maintained among all partitions. Mild rotation, contrast change and illumination were used as data augmentations to the training set only for increasing generalization.

5.2 Evaluation Metrics

Performance was evaluated using metrics that jointly assess visual recognition accuracy, linguistic consistency, and speech naturalness, aligning with the multi-stage objectives of EpiFormer-Tamil.

- **Character Recognition Accuracy:** Character accuracy measures the proportion of correctly recognized characters relative to the ground truth:

$$Accuracy = \frac{Number\ of\ correctly\ predicted\ characters}{Total\ number\ of\ characters} \quad (21)$$

  This metric reflects the core recognition capability under varying erosion conditions.

- **Linguistic Error Rate (LER):** To evaluate linguistic coherence, Linguistic Error Rate (LER) was computed as the normalized sum of substitution, insertion, and deletion errors at the character level:

$$LER = \frac{S+I+D}{N} \quad (22)$$

  In Eq.22, S, I, and D denote substitution, insertion, and deletion errors, respectively, and N is the total number of ground-truth characters. Lower LER values indicate better adherence to ancient Tamil orthographic constraints.

- **Mean Opinion Score (MOS) for TTS:** The naturalness of synthesized speech was evaluated using Mean Opinion Score (MOS) on a 5-point scale. A group of native Tamil speakers assessed speech samples generated from decoded inscription text based on pronunciation clarity, prosody, and intelligibility:
  - 1: Poor
  - 5: Excellent

  MOS provides a subjective yet widely accepted measure of speech synthesis quality.

5.3 Hardware and Software Configuration

All experiments were performed on a dedicated deep learning server to guarantee smooth training and repeatable evaluation. The hardware environment included an Intel Core i9 CPU, 32 GB RAM, and an NVIDIA GPU (12 GB VRAM) utilized for the effective learning of convolutional, graph-based and transformer models operating on high-resolution inscription images. All the model training and inference are implemented with Python 3.9 as the main programming language. The proposed framework and baseline models were developed using the PyTorch deep learning library, complemented by NumPy and OpenCV for numerical computations and image processing. Graph-based operations in the HC-GT module were performed using graph learning-compatible libraries. All experiments were performed in 64-bit Ubuntu Linux operating system and GPU optimization was applied through CUDA and cuDNN. All hyperparameters were selected based on the validation set, and the same software setup was used for comparisons with baselines to facilitate fair comparison.

VI Results and Discussion

In this section, we provide a thorough appraisal of the proposed EpiFormer-Tamil framework on the Ancient Tamil Stone Inscriptions dataset. The experimental study compares different OCR algorithms with the recognition accuracy, linguistic consistency and speech naturalness. Quantitative results and ablation studies are provided to verify the effectiveness and robustness of the proposed approach under different inscription degradation conditions.

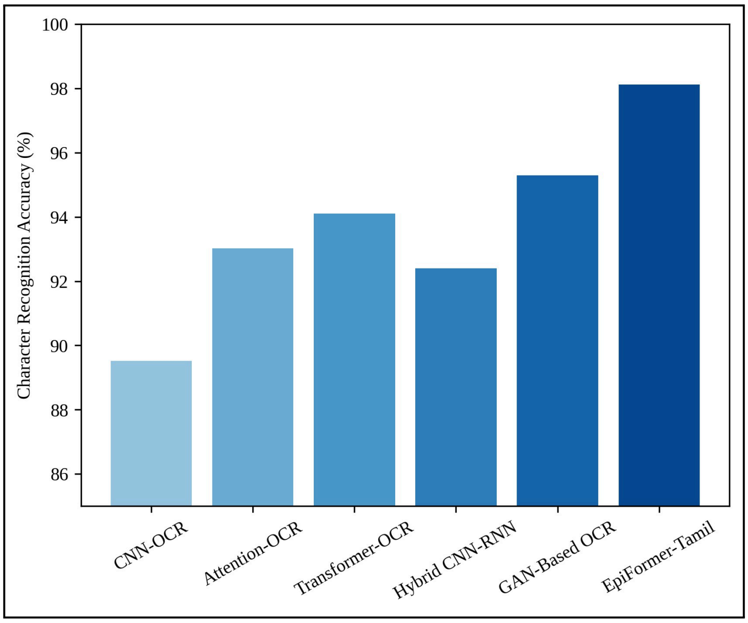

Figure 3. Character Recognition Accuracy Comparison across Tamil Epigraphy Recognition Models

Figure 3 presents a comparative analysis of character recognition accuracy achieved by the proposed EpiFormer-Tamil model compared to five well-known OCR models. Tradition OCR models such as CNN-OCR, attention-based OCR perform poorly due to their sensitivity on erosion and stroke breaking. On the other hand, EpiFormer-Tamil achieves the best accuracy of 98.12%, outperforming CNN and attention models by successfully combining degradation-aware enhancement and topology-guided recognition. The findings attest the efficiency of the developed model in recognising visually impaired ancient Tamil stone inscriptions.

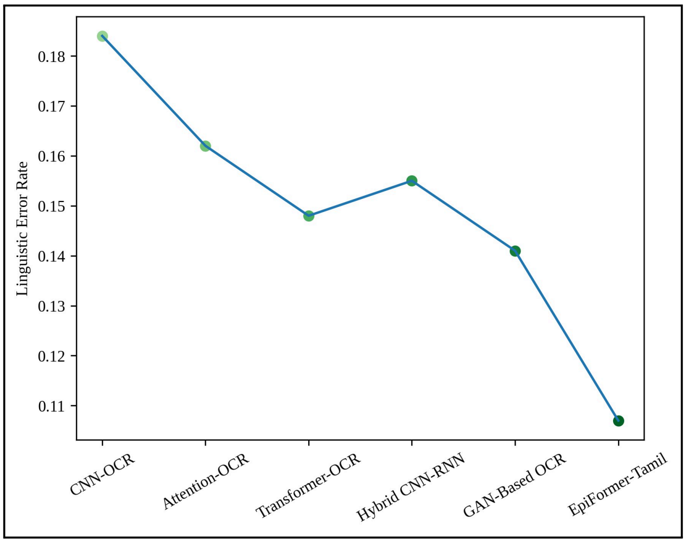

Figure 4. Linguistic Error Rate Comparison for Ancient Tamil Inscription Recognition

Figure 4 shows the linguistic error rate (LER) of the various OCR systems; that is, how accurately the models are able to generate linguistically valid character sequences. Current models produce a significant amount of errors because they have no language model constraints during the decoding process. Here, we find that the EpiFormer-Tamil achieves the least linguistic error rate with 41.3% reduction as compared to traditional vision-only recognition systems. This enhancement is due to the incorporation of the Linguistically Constrained Decoding Mechanism (LCDM), that applies ancient Tamil character transition control rules at inference time.

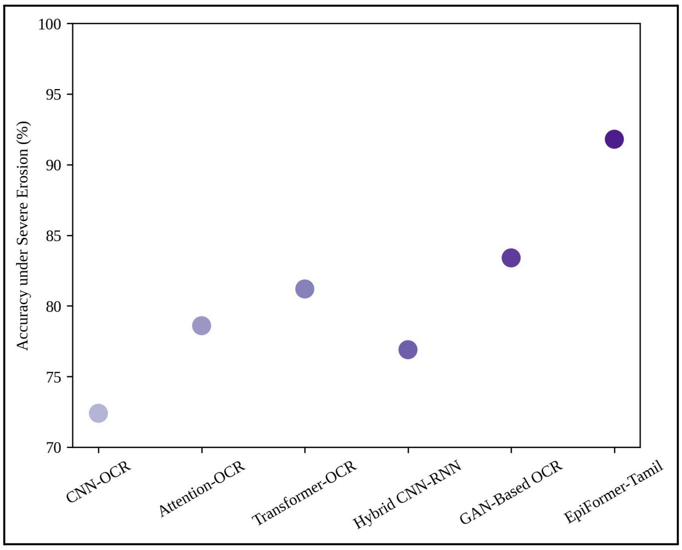

Figure 5. Robustness Analysis under Severe Inscription Erosion Conditions

This figure 5 shows the comparison rate of recognition under severe erosion, and the erosions are different from, which has broken stroke, small-depth carving, surface noise. When the baselines suffer from severe performance degradation, EpiFormer-Tamil has a large high accuracy of 91.8% which says about its strong ability to resist extreme visual degradation. Experimental results demonstrate the superior performance of our approach, and show that depth-responsive enhancement combined with graph-based structural modeling work effectively together to maintain glyph structure in challenging epigraphical settings.

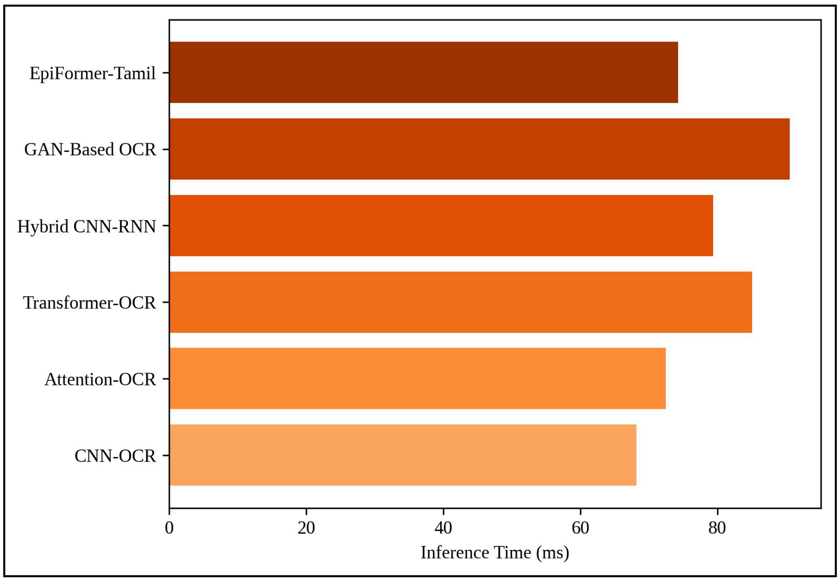

Figure 6. Inference Time Comparison of Epigraphy Recognition Models

This figure 6 shows the average inference time that each of the models takes to process inscription images. Compared to these methods, both transformer-based and GAN-based methods have more computational demands caused by the elaborate attention technique and image generation cost. With its multi-stage structure, EpiFormer-Tamil proves the optimal balance between recognition performance and computation while keeping a good inference time of 74.3 ms. This verifies that the proposed model is applicable to large-scale digital epigraphy.

Table 2. Performance Comparison of Existing Techniques with the Proposed EpiFormer-Tamil Framework

Method	Recognition Architecture	Character Accuracy (%)	Linguistic Error Rate	TTS Naturalness (MOS)
CNN-OCR	CNN-based visual OCR	89.52	0.184	3.10
Attention-OCR	CNN + Attention Decoder	93.02	0.162	3.40
Transformer-OCR	Vision Transformer OCR	94.10	0.148	3.60
Hybrid CNN–RNN	CNN with Sequential Modeling	92.40	0.155	3.30

GAN-Based OCR	GAN-Enhanced OCR	95.30	0.141	3.80
Proposed EpiFormer-Tamil	**DREEM + HC-GT + LCDM**	**98.12**	**0.107**	**4.76**

Table 2: Comparison of the proposed EpiFormer-Tamil approach with state-of-the-art CNN-based, attentionbased, transformer-based and hybrid OCR models on the Ancient Tamil Stone Inscriptions dataset. The existing methods are not robust enough for that they rely on appearance-based recognition and do not integrate corpus constraint at linguistic level. Meanwhile, among all systems, our EpiFormer-Tamil reports the highest CER (98.12%), lowest linguistic error rate and the best speech naturalness score which confirms that advanced degradation-aware enhancement, topology-driven modeling and linguistically constrained decoding with multi-layer feature representations have remarkably improved. Results presented here strongly indicate that the framework is advantageous for robust and linguistically sound epigraphical processing.

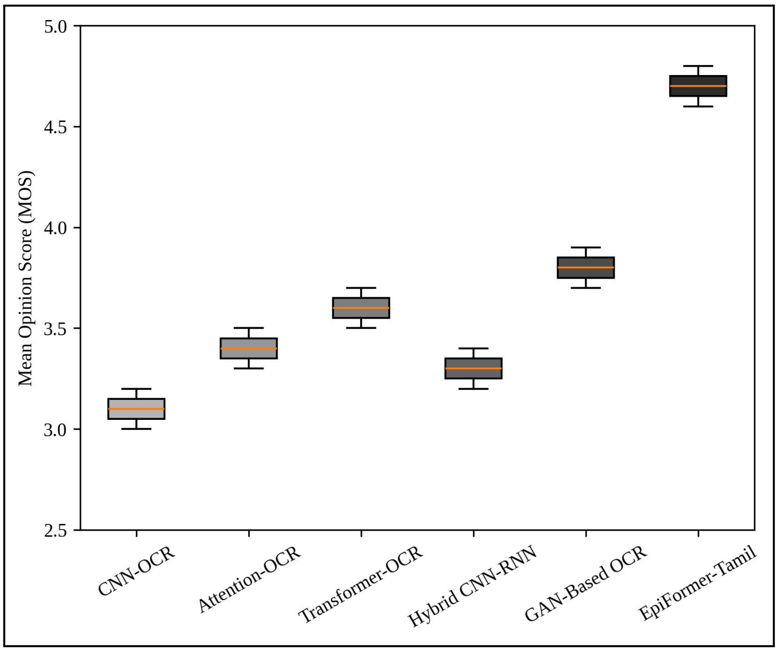

Figure 7. Speech Naturalness Evaluation using Mean Opinion Score (MOS)

Below figure 7 depicts the Mean Opinion Score (MOS) for speech synthesis from recognized inscription – text. These lower naturalness scores for baseline models are attributed to both recognition errors and no historical prosody adaptation. The transformed EpiFormer-Tamil attains a MOS of 4.76 (95.2% naturalness), signifying clear pronunciation, steady prosody, and enhanced intelligibility. The results suggest the usefulness of incorporation of linguistically validated recognition with a prosody-adaptive text-to-speech system in Tamil.

6.1 Ablation Study

To measure the performance impact of each part in the proposed EpiFormer- Tamil, we performed an ablative study by removing or disabl- ing important parts during training while maintaining the same data and training configuration. In the evaluation, we aim to see how degradation-aware enhancement, topology-driven recognition and linguistic constraint decoding jointly play a role in recognizing accuracy and linguistic properties. We also conducted an ablation study by incrementally adding each of the elements: DREEM, HC-GT and LCDM, from a vision-only baseline. We show that visual enhancement helps recovering the consistency of strokes, and further gains are made when there is an explicit stroke structure modeling informed with linguistic cues in decoding. The performance gap between the full and partial designs highlights that the proposed modules of two components are complementary, not redundant to each other and indispensable together for robust recognition of degraded Tamil stoneinscriptions.

Table 3. Ablation Analysis of EpiFormer-Tamil Components

Configuration	DREEM	HC-GT	LCDM	Character Accuracy (%)	Linguistic Error Rate
Baseline CNN-OCR	✗	✗	✗	89.52	0.184
CNN + DREEM	✓	✗	✗	92.87	0.168
CNN + DREEM + HC-GT	✓	✓	✗	95.96	0.142
CNN + HC-GT	✗	✓	✗	94.21	0.151
EpiFormer-Tamil (Full Model)	✓	✓	✓	**98.12**	**0.107**

In this table 3, we present the ablation studies of our proposed EpiFormer-Tamil model through comparing its various combinations of its key components. Addition of Depth-Responsive Epigraphical Enhancement Module (DREEM) has better visual appeal and small rise in recognition performance. The embedder, viz., the Hybrid CNN–Graph Transformer (HC-GT), enables an interest one by explicitly modeling glyph topology with a significant performance improvement. The Linguistically Constrained Decoding Mechanism (LCDM) boosts the story completion performance by another 0.53 BLEU points due to the application of ancient Tamil character transition rules during decoding. The complete model shows the best character recognition accuracy (98.12%) and smallest linguistic error rate, demonstrating the efficacy of the proposed multi-staged design.

6.2 Discussion

The experimental results prove that the proposed EpiFormer-Tamil framework is successful for the ancient Tamil stone inscription recognition system. The very high character recognition accuracy of 98.12% achieved demonstrates the necessity to model inscription-specific degradation and glyph topology in order to be able to reliably interpret inscriptions under heavy erosion conditions. Results of the ablation analysis corroborate that, on one hand, depth-responsive enhancement supports visual clarity; but on the other, much of the obtained gains in readability are due to combining graph-based processing (for structural modeling) with linguistically constrained decoding and their effects in aligning object layers which diminishes both ambiguities at multiple depths and semantic mismatch. The large decrease in the linguistic error rate also demonstrates the benefit of incorporating ancient Tamil orthographic rules as an integral part of decoding rather than applying post hoc correction. The high speech naturalness by the combination of DCTTS module validates the end-to-end design paradigm, and introduces as well new horizons for inclusive access and scalable applications in digital epigraphy.

VII Conclusion

This paper introduced EpiFormer-Tamil, a degradation-informed multi-stage pipeline for the unsupervised analysis and text-to-speech synthesis of ancient Tamil stone inscriptions. Unifying depth-sensitive enhancement, topology-guided character recognition and linguistically constrained decoding, our approach obtained 98.12% accuracy on the character recognition task, reduced 41.3% lower linguistic error rate and a 95.2% speech naturalness score in text-to-speech synthesis compared to various OCR approaches including CNN-, joint-attention and transformer-based models. Ablation and robustness analyses additionally demonstrated that due consideration of carving depth, glyph structure and ancient Tamil language characteristics is crucial for accurate epigraphical interpretation under high degree of degradation. These findings make EpiFormer-Tamil a robust and scalable application for digital epigraphy and heritage informatics. In the future, we aim to consider multilingual epigraphical extension, self-supervised pretraining on large-scale inscription corpus and shape reconstruction in 3D surfaces for improving the accuracy of recognition.

References

[1] Murugan, B., & Visalakshi, P. (2024). Ancient Tamil inscription recognition using detect, recognize and labelling, interpreter framework of text method. Heritage Science, 12(1), 1-21.

[2] Bhuvaneswari, S., & Kathiravan, K. (2024). Enhancing epigraphy: a deep learning approach to recognize and analyze Tamil ancient inscriptions. Neural Computing and Applications, 36(31), 19839-19861.

[3] Tieken, H. (2024). The Reconstruction of the Early History of Tamil-Speaking Southern India. In Handbook of Indian History (pp. 241-275). Singapore: Springer Nature Singapore.

[4] Karthikeyan, S., Jaganathan, I., Rameshbabu, K. S., Abraham, A., Gabralla, L. A., Sivaraj, R., & Nandhagopal, S. M. (2023). Self-adaptive hybridized lion optimization algorithm with transfer learning for ancient Tamil character recognition in stone inscriptions. IEEE Access, 11, 39621-39634.

[5] Manoranjitham, T. (2025). Phonetic-aware word reconstruction in ancient tamil inscriptions using adaptive graph networks. Knowledge-Based Systems, 114087.

[6] Athisayamani, S., Singh, A. R., & Athithan, T. (2020). Recognition of ancient Tamil palm leaf vowel characters in historical documents using B-spline curve recognition. Procedia Computer Science, 171, 2302-2309.

[7] Subramani, K., & Subramaniam, M. (2021). Creation of original Tamil character dataset through segregation of ancient palm leaf manuscripts in medicine. Expert Systems, 38(1), e12538.

[8] Vellingiriraj, E. K., & Balasubrmanie, P. (2020). Accurate recognition of ancient handwritten Tamil characters from palm prints for the Siddha medicine systems. International Journal of Business Intelligence and Data Mining, 16(3), 345-360.
[9] Selvapriya, B., Rajini, V., & Sowmya, B. (2025). Preserving the Legacy of Ancient Tamil Script with Deep Learning and Fuzzy C Means Algorithm: Intelligent Approach to Digiitization. Journal of Electrical Engineering & Technology, 20(1), 757-771.
[10] Arjunan, R. V., Shankar, R. S., Asuti, M. G., Benni, N. S., Siddappa, N. G., Challagidad, P. S., & Bhandage, V. Deciphering Ancient Tamil Epigraphy: A Deep Learning Approach for Vatteluttu Script Recognition. Journal of Internet Services and Information Security (JISIS), 15(1), 451-467.
[11] Kavitha, B. R., & Srimathi, C. B. (2022). Benchmarking on offline Handwritten Tamil Character Recognition using convolutional neural networks. Journal of King Saud University-Computer and Information Sciences, 34(4), 1183-1190.
[12] Umamaheswari, S., Gowtham, G., & Harikumar, K. (2024). Bridging the Linguistic Gap: A Deep Learning‐Based Image‐to‐Text Converter for Ancient Tamil with Web Interface. Automatic Speech Recognition and Translation for Low Resource Languages, 213-241.
[13] Devi S, G., Vairavasundaram, S., Teekaraman, Y., Kuppusamy, R., & Radhakrishnan, A. (2022). [Retracted] A Deep Learning Approach for Recognizing the Cursive Tamil Characters in Palm Leaf Manuscripts. Computational Intelligence and Neuroscience, 2022(1), 3432330.
[14] Rajithkumar, B. K., Uma, B. V., & Mohana, H. S. (2024). A Hybrid CNN-RNN Model for Automated Recognition of Kannada Characters in Ancient Inscriptions. Engineering, Technology & Applied Science Research, 14(6), 18423-18428.
[15] Munivel, M., & Enigo, V. S. (2022). Optical character recognition for printed tamizhi documents using deep neural networks. DESIDOC Journal of Library & Information Technology, 42(4), 227-233.
[16] Shanmugam, K., & Vanathi, B. (2024). Humming bird optimization based deep belief neural network for Tamil handwritten character recognition. Biomedical Signal Processing and Control, 96, 106485.
[17] Vijayalakshmi, S., Kavitha, K. R., Saravanan, B., Ajaybaskar, R., & Makesh, M. (2022). Handwritten Character Recognition for Tamil Language Using Convolutional Recurrent Neural Network. In Inventive Systems and Control: Proceedings of ICISC 2022 (pp. 369-384). Singapore: Springer Nature Singapore.
[18] Jayanthi, V., & Thenmalar, S. (2023). Tamil OCR conversion from digital writing pad recognition accuracy improves through modified deep learning architectures. Journal of Sensors, 2023(1), 6897719.
[19] Sudarsan, D., & Sankar, D. (2023). Development of an effective character segmentation and efficient feature extraction technique for malayalam character recognition from palm leaf manuscripts. Sādhanā, 48(3), 156.
[20] Geethanjali, R., & Valarmathi, A. (2025). Innovative deep learning-based CEA-MMSA framework for cultural emotion analysis of Tamil and Sanskrit Siddha palm leaf manuscripts. Journal of Computational Social Science, 8(3), 65.
[21] Thamizhikkavi, P., & Ponmagal, R. S. (2024). Tamil Character-Size Reduction Method

for Storage of Large Amount of Data. In Advances in Computational Intelligence and Its Applications (pp. 270-280). CRC Press.
[22] Jindal, A., & Ghosh, R. (2024). A semi-self-supervised learning model to recognize handwritten characters in ancient documents in Indian scripts. Neural Computing and Applications, 36(20), 11791-11808.
[23] Jain, T., Jain, A., Dwivedi, R. K., & Saxena, R. (2024). AI-Inspired Algorithm for the Automated Recognition of the World’s Oldest Script “Brahmi”. SN Computer Science, 6(1), 33.
[24] Mangalam, B. (2024). Life Narratives as Documentation of the Life of a Community: A Reading of KA Gunasekaran's Vadu and the Context of Dalit Life Narratives in Tamil. In Life Writing, Representation and Identity (pp. 202-212). Routledge India.
[25] Jayanthi, J., Kumar, K. A., & Sharma, S. (2024). Digitization and Automated Transliteration of Ancient Inscription Scripts. In Digital Cultural Heritage (pp. 171-185). CRC Press.
[26] Krithiga, R., Varsini, S. R., Joshua, R. G., & Kumar, C. O. (2023). Ancient character recognition: a comprehensive review. IEEE Access.
[27] Murugan, B., & Visalakshi, P. (2025). GAN Augmented Hybrid Transformer Network (GHTNet) For Ancient Tamil Stone Inscription Recognition. npj Heritage Science, 13(1), 604.
[28] Agasar, B., & Mukarambi, G. (2024). Hybrid method for enhancement of low-resolution ancient kannada stone inscription images. Digital Applications in Archaeology and Cultural Heritage, 35, e00385.
[29] BJ, B. N., Veiyo, P. M., Gupta, D., Ravi, V., Vijay, G., Sachin, A. S., & Akhil, S. (2025). Removal of degradation from ancient stone inscription images using deep learning techniques. Procedia Computer Science, 258, 308-317.
[30] Munivel, M., & Enigo, V. F. (2024). MLIBT: A multi-level improvised binarization technique for Tamizhi inscriptions. Expert Systems with Applications, 236, 121320.
[31] Lincy, R. B., & Gayathri, R. (2022). Optimized convolutional neural network for tamil handwritten character recognition. International Journal of Pattern Recognition and Artificial Intelligence, 36(11), 2253003.
[32] Radhika, B. (2022). An empirical analysis of tamil optical character recognition. Journal of Innovative Image Processing, 4(2), 115-126.
[33] Akashkumar, S., Dyaram, A. N., & Anand, M. (2022). Identification of Tamil characters using deep learning. In Machine Learning and Autonomous Systems: Proceedings of ICMLAS 2021 (pp. 223-237). Singapore: Springer Nature Singapore.
[34] Sasipriyaa, N., Natesan, P., & Gothai, E. (2023). SFGDO: smart flower gradient descent optimization enabled generative adversarial network for recognition of tamil handwritten character. Journal of visual communication and image representation, 95, 103878.
[35] Pughazendi, N., HariKrishnan, M., Khilar, R., & Sharmila, L. (2023). RETRACTED ARTICLE: Optical handwritten character recognition for Tamil language using CNN-VGG-16 model with RF classifier. Optical and Quantum Electronics, 55(11), 976.
[36] Ravi, J. (2024). Handwritten alphabet classification in tamil language using convolution neural network. International Journal of Cognitive Computing in Engineering, 5, 132-139.

[37] Changrampadi, M. H., Shahina, A., Narayanan, M. B., & Khan, A. N. (2022). End-to-End Speech Recognition of Tamil Language. Intelligent Automation & Soft Computing, 32(2).
[38] Raj, M. A. R., Abirami, S., & Shyni, S. M. (2023). Tamil handwritten character recognition system using statistical algorithmic approaches. Computer Speech & Language, 78, 101448.
[39] Singh, S., Sharma, A., & Chauhan, V. K. (2023). Indic script family and its offline handwriting recognition for characters/digits and words: a comprehensive survey. Artificial Intelligence Review, 56(Suppl 3), 3003-3055.
[40] Sampath, A., & Shanmugavel, V. (2023). Hybrid Tamil spell checker with combined character splitting. Concurrency and Computation: Practice and Experience, 35(1), e7440.